

A Prospective Study of Prepregnancy Dietary Iron Intake and Risk for Gestational Diabetes Mellitus



OBJECTIVE It is important to identify modifiable factors that may lower gestational diabetes mellitus (GDM) risk. Dietary iron is of particular interest given that iron is a strong prooxidant, and high body iron levels can damage pancreatic β -cell function and impair glucose metabolism. The current study is to determine if prepregnancy dietary and supplemental iron intakes are associated with the risk of GDM. **RESEARCH DESIGN AND METHODS** A prospective study was conducted among 13,475 women who reported a singleton pregnancy between 1991 and 2001 in the Nurses' Health Study II. A total of 867 incident GDM cases were reported. Pooled logistic regression was used to estimate the relative risk (RR) of GDM by quintiles of iron intake controlling for dietary and nondietary risk factors. **RESULTS** Dietary heme iron intake was positively and significantly associated with GDM risk. After adjusting for age, BMI, and other risk factors, RRs (95% CIs) across increasing quintiles of heme iron were 1.0 (reference), 1.11 (0.87–1.43), 1.31 (1.03–1.68), 1.51 (1.17–1.93), and 1.58 (1.21–2.08), respectively (P for linear trend 0.0001). The multivariate adjusted RR for GDM associated with every 0.5-mg per day of increase in intake was 1.22 (1.10–1.36). No significant associations were observed between total dietary, nonheme, or supplemental iron intake and GDM risk. **CONCLUSIONS** These findings suggest that higher prepregnancy intake of dietary heme iron is associated with an increased GDM risk.